Arithmetic-Geometric Structure of Agrawal's Conjecture and the Torus Dimension Count Behind the Double Condition

(AG track working document — v0.2, items AG2 and AG5)

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Abstract

We reframe Agrawal's conjecture and the Lenstra–Pomerance counterexample mechanism in the language of one-dimensional algebraic groups. The main new result (item AG2 of `doc/arith-geom-plan.md`) upgrades the empirical finding A11 of this project (`data/FINDINGS.md`) to a structural statement: for *every* AKS modulus r, the constructive (Lenstra–Pomerance-style) route forces smoothness control on the $\mathbb{F}_p$-point orders of a product of cyclotomic tori of total dimension at least 2, and the minimum 2 is attained exactly by the split and nonsplit one-dimensional tori — orders $p-1$ and $p+1$ — i.e. by the Carmichael + Lucas–Carmichael double condition (equivalently, twin-smoothness). The key identity is an inversion collapse: when $d = \mathrm{ord}_r(p)$ is even, the unique involution of $(\mathbb{Z}/r)^\times$ is -1, forcing $(X-1)^{p^{d/2}-1} \equiv -X^{-1} \pmod{p,\, \Phi_r(X)}$ and collapsing the order of $X-1$ into the cyclotomic levels $e \mid d/2$. Since $\sum_{e\mid m} \varphi(e) = m$, the burden is $d/2$ for even d and d for odd d; validity of a counterexample forces $d \geq 3$, so the burden is minimized at $d = 4$ (possible iff $4 \mid r-1$), value 2, never 1. All identities and the resulting burden table are verified numerically (`code/ag2_torus_check.py`), reproducing A11's measured table for all eleven moduli tested there. Section 5 (item AG5) extends the count to Popovych's two-section conjecture: the second section $X+2$ has trivial stabilizer in the Galois action — its zero -2, unlike the zero 1 of $X-1$, is not inversion-stable — so no collapse exists at any depth, the burden rises to $\min\{d : d \mid r-1,\ d \geq 3\} \geq 3$ for every modulus (4 at Popovych's own $r = 5$: $p-1$, $p+1$ and p^2+1 simultaneously smooth), and the twin-smooth regime, the only one this project found supply-feasible, does not exist for the pair. This is the first quantified evidence in either direction on Popovych's conjecture beyond search ranges.

1 The Frobenius-lift dictionary

This section fixes the geometric reading used throughout; full exposition is item AG1 and will accrete here. Write $T(-1, n, r)$ for the AKS congruence

$$(X-1)^n \equiv X^n - 1 \pmod{n,\, X^r - 1}, \tag{1}$$

r an odd prime not dividing n.

Proposition 1.1 (primality as a Frobenius lift)**.** *Let $A_n = (\mathbb{Z}/n)[X]/(X^r-1)$, the mod-$n$ group algebra of μ_r, and let $\sigma_n : X \mapsto X^n$ be the n-th Adams operation (well defined since $X^r = 1$). If n is prime, then the n-power map on A_n coincides with σ_n; that is, $f(X)^n \equiv f(X^n) \pmod{n,\, X^r-1}$ for* every *f. The congruence (1) is this coincidence tested on the single element $f = X-1$.*

Proof. For n prime the n-power map is a ring endomorphism (freshman's dream) and fixes $\mathbb{Z}/n$ (Fermat), so $f(X)^n \equiv f^{(n)}(X^n) \equiv f(X^n)$, where $f^{(n)}$ raises coefficients to the n-th power. □

In λ-ring language: $\mathbb{Z}[X]/(X^r-1) = \mathbb{Z}[\mathbb{Z}/r]$ carries the Adams operations $\psi^k(X) = X^k$, and "n is prime" is exactly "ψ^n is a Frobenius lift at n" (Wilkerson's criterion direction); (1) is the lift condition at one section of μ_r. **[TODO: AG1: cite Wilkerson, Borger, and the pseudofield formalism of Lenstra–Pomerance.]**

Lemma 1.2 (the exceptional clause)**.** $n^2 \equiv 1 \pmod r$ *iff* σ_n*, viewed as the element* $n \bmod r$ *of* $(\mathbb{Z}/r)^\times \cong \mathrm{Gal}(\mathbb{Q}(\zeta_r)/\mathbb{Q})$*, lies in* $\mathrm{Gal}(\mathbb{Q}(\zeta_r)/\mathbb{Q}(\zeta_r)^+) = \{1, \textit{complex conjugation}\}$.

Proof. $(\mathbb{Z}/r)^\times$ is cyclic of even order $r-1$; its unique element of order 2 is -1, which acts on μ_r as inversion, i.e. as complex conjugation, the generator of $\mathrm{Gal}(\mathbb{Q}(\zeta_r)/\mathbb{Q}(\zeta_r)^+)$. □

Agrawal's conjecture thus asserts: *a composite Frobenius impostor on the single section* $X-1$ *must act trivially on the maximal real subfield.* A *valid counterexample* is a composite n with $T(-1,n,r)$ and $t := \mathrm{ord}_r(n) > 2$.

2 Setup: cyclotomic tori and the mechanism

2.1 Cyclotomic tori

For $e \geq 1$ let Φ_e denote the e-th cyclotomic polynomial, so that

$$p^d - 1 \;=\; \prod_{e \mid d} \Phi_e(p). \tag{2}$$

Over $\mathbb{F}_p$, the Weil restriction $\mathrm{Res}_{\mathbb{F}_{p^d}/\mathbb{F}_p}\mathbb{G}_m$ decomposes up to isogeny as $\prod_{e|d} T_e$, where T_e is the cyclotomic torus of dimension $\varphi(e)$ split by $\mathbb{F}_{p^e}$, with $\#T_e(\mathbb{F}_p) = \Phi_e(p)$ [8, 9]. The two *one-dimensional* tori are $T_1 = \mathbb{G}_m$ ($\#T_1(\mathbb{F}_p) = p-1$, the split torus) and T_2 the norm-one torus of $\mathbb{F}_{p^2}/\mathbb{F}_p$ ($\#T_2(\mathbb{F}_p) = p+1$, the nonsplit twist); $\varphi(e) = 1$ iff $e \in \{1,2\}$. We refer to the index e in (2) as the *level*.

2.2 The order of $X-1$

Fix primes $p \neq r$, $p \nmid n$, and let $d = d_p := \mathrm{ord}_r(p)$. Then

$$\mathbb{F}_p[X]/(X^r-1) \;\cong\; \mathbb{F}_p \times \mathbb{F}_p[X]/(\Phi_r(X)), \qquad \mathbb{F}_p[X]/(\Phi_r(X)) \;\cong\; \prod_{i=1}^{(r-1)/d} \mathbb{F}_{p^d},$$

the factors indexed by the orbits of $\langle p\rangle$ on $(\mathbb{Z}/r)^\times$; in the i-th factor X maps to a primitive r-th root of unity ζ_i. The element $X-1$ vanishes in the $\mathbb{F}_p$ factor (where (1) holds trivially: both sides are 0) and is a unit in $\mathbb{F}_p[X]/\Phi_r$ since $\Phi_r(1) = r \neq 0$ in $\mathbb{F}_p$.

Definition 2.1. $\rho_r(p)$ *is the multiplicative order of the image of* $X-1$ *in* $\big(\mathbb{F}_p[X]/\Phi_r(X)\big)^\times$*, i.e.* $\rho_r(p) = \mathrm{lcm}_i \,\mathrm{ord}(\zeta_i - 1)$.

For $r = 5$ this is the quantity $\rho(p)$ of Váňa [7], with $\rho(p) \mid 10(p^2-1)$; Corollary 3.3 below recovers this.

2.3 The Lenstra–Pomerance mechanism

Definition 2.2 (L–P mechanism)**.** *A squarefree composite $n = p_1 \cdots p_k$ coprime to r is produced by the* L–P mechanism *if for every i there is $j_i \geq 0$ with*

$$n \equiv p_i^{j_i} \pmod{\operatorname{lcm}\big(r,\ \rho_r(p_i)\big)}, \tag{3}$$

and $\operatorname{ord}_r(n) > 2$ (validity).

The original $r = 5$ construction [2, 4] is the instance $j_i = 1$: the conditions $p \equiv 3 \pmod{80}$, $p - 1 \mid n - 1$, $p + 1 \mid n + 1$ give $n \equiv p \pmod{\operatorname{lcm}(5, \rho_5(p))}$ once $\rho_5(p) \mid 10(p^2 - 1)$ is known. Our entire Track A pipeline (`PLAN.md`) is a search inside Definition 2.2.

Remark 2.3 (scope). The mechanism is sufficient, not necessary, for (1): the component equation $(\zeta - 1)^n = \zeta^n - 1$ could conceivably hold "accidentally" for some p, outside the Frobenius-matching family (3). No such solutions are known, and no way to construct them is known; every published counterexample construction [2, 4, 6] is an instance of Definition 2.2. All results below quantify *this mechanism*, in line with the scope discipline of the negative-result paper (`doc/negative-result-plan.md`).

3 The torus dimension count

3.1 Frobenius matching and the inversion collapse

Lemma 3.1 (Frobenius matching)**.** *With notation as above:*

(i) $(X - 1)^{p^k} \equiv X^{p^k} - 1 \pmod{p,\ X^r - 1}$ *for all $k \geq 0$.*

(ii) If $n \equiv p^j \pmod{\operatorname{lcm}(r, \rho_r(p))}$ *then $T(-1, n, r)$ holds modulo p.*

(iii) Conversely, (3) *forces* $n \equiv p^j \pmod{r}$*, hence* $\operatorname{ord}_r(n) \mid d_p$*. In particular a valid counterexample (*$\operatorname{ord}_r(n) > 2$*) forces $d_p \geq 3$ for every $p \mid n$.*

Proof. (i) is the iterated Frobenius endomorphism of $\mathbb{F}_p$-algebras. (ii): write $n = p^j + c \cdot \operatorname{lcm}(r, \rho_r(p))$. In the $\mathbb{F}_p$ factor both sides of (1) vanish. In $\mathbb{F}_p[X]/\Phi_r$: $(X-1)^n = (X-1)^{p^j}\big((X-1)^{\rho_r(p)}\big)^{c'} = (X - 1)^{p^j} \equiv X^{p^j} - 1$ by (i), and $X^{p^j} = X^{n \bmod r} = X^n$ since $X^r \equiv 1 \bmod \Phi_r$ (as $X^r - 1 = (X-1)\Phi_r(X)$ for r prime). (iii): reduce (3) modulo r; then $n \in \langle p \rangle \subseteq (\mathbb{Z}/r)^\times$, so $\operatorname{ord}_r(n)$ divides $\#\langle p \rangle = d_p$, and $\operatorname{ord}_r(n) \geq 3$ gives $d_p \geq 3$. □

Lemma 3.2 (inversion collapse)**.** *Suppose $d = \operatorname{ord}_r(p)$ is even. Then, in $\mathbb{F}_p[X]/\Phi_r(X)$,*

$$(X - 1)^{p^{d/2} - 1} \ = \ -X^{-1},$$

and consequently

$$\rho_r(p) \ \Big| \ 2r\,(p^{d/2} - 1) \ = \ 2r \prod_{e \mid d/2} \Phi_e(p). \tag{4}$$

Proof. $(\mathbb{Z}/r)^\times$ is cyclic, so its unique element of order 2 is -1; since $\operatorname{ord}_r(p) = d$, the element $p^{d/2}$ has order 2, whence $p^{d/2} \equiv -1 \pmod{r}$. In each field component, with $X \mapsto \zeta$ a primitive r-th root of unity, iterated Frobenius gives

$$(\zeta - 1)^{p^{d/2}} = \zeta^{p^{d/2}} - 1 = \zeta^{-1} - 1 = -\zeta^{-1}(\zeta - 1),$$

and dividing by $\zeta - 1 \neq 0$ yields $(\zeta-1)^{p^{d/2}-1} = -\zeta^{-1}$. This holds in every component, hence in the product ring. Finally $(-X^{-1})^{2r} = X^{-2r} = 1$, so $(X-1)^{2r(p^{d/2}-1)} = 1$, giving (4); the product form is (2) applied to $p^{d/2}-1$. □

Corollary 3.3 (Váňa's bound)**.** *For $r = 5$ and $p \equiv \pm 2 \pmod 5$ (so $d = 4$), $\rho_5(p) \mid 10(p^2-1)$.*

Definition 3.4 (support level)**.** *Set*

$$d' \;=\; d'(p) \;:=\; \begin{cases} d/2, & d \text{ even}, \\ d, & d \text{ odd}. \end{cases}$$

For odd d the trivial bound $\rho_r(p) \mid p^d - 1 = \prod_{e \mid d} \Phi_e(p)$ applies (orders in a product of copies of $\mathbb{F}_{p^d}^{\times}$), so in both cases

$$\rho_r(p) \;\Big|\; 2r \prod_{e \mid d'} \Phi_e(p)\,: \tag{5}$$

the p-dependent part of $\rho_r(p)$ is supported on the cyclotomic levels $e \mid d'$.

3.2 Generic sharpness

The bound (5) is an upper bound on support; the mechanism's cost analysis also needs it to be generically *attained* — otherwise a harvest could fish for primes p where $\rho_r(p)$ collapses further.

Proposition 3.5 (generic sharpness; heuristic + empirical)**.** *Let $\ell > 2rd$ be a prime with $\ell \mid \Phi_e(p)$ for some $e \mid d'$. Then $\ell \mid \rho_r(p)$ unless in every component the image of $X-1$ is an ℓ-th power in $\mathbb{F}_{p^d}^{\times}$. Modelling the components as independent uniform elements, the exceptional probability is $\ell^{-(r-1)/d} \leq 1/\ell$. Consequently, for all p outside a set of (heuristic) density $O(\sum_\ell 1/\ell)$, the large-prime part of $\prod_{e \mid d'} \Phi_e(p)$ divides $\rho_r(p)$, and requiring "$\rho_r(p)$ is B-smooth" is equivalent to requiring "$\Phi_e(p)$ is B-smooth for every $e \mid d'$".*

Evidence. $\ell \nmid \rho_r(p)$ iff the image of $X-1$ has trivial ℓ-part in each cyclic component group $\mathbb{F}_{p^d}^{\times}$, i.e. is an ℓ-th power in every component whose order ℓ divides; the ℓ-th powers have index ℓ in each component containing ℓ-torsion. Rigorously controlling the image of $\zeta - 1$ is an open Artin-type problem; empirically (`code/ag2_torus_check.py`, 67 sample primes across $r \in \{5, 7, 11, 13, 19, 23, 41\}$, $d \in \{3, 4, 5, 6, 11\}$): all 37 tested large primes ℓ divided $\rho_r(p)$ (0 exceptions; ≈ 0.19 expected under the $1/\ell$ model), and in all 67 cases removing the top level $e = d'$ from the bound (5) broke it (the top level is genuinely present in $\rho_r(p)$). □

3.3 The dimension-count theorem

Theorem 3.6 (torus dimension count)**.** *Let r be an odd prime and let n be produced by the L–P mechanism (Definition 2.2) with $\mathrm{ord}_r(n) > 2$. For each $p \mid n$ put $d_p = \mathrm{ord}_r(p)$ and d'_p as in Definition 3.4. Assume generic sharpness (Proposition 3.5) at p. Then, for the CRT system (3) to admit assembly over many primes with a common B-smooth modulus (the setting of the Track-A pipeline), each $p \mid n$ must satisfy:*

$$\Phi_e(p) \text{ is } B\text{-smooth for every } e \mid d'_p \quad\Longleftrightarrow\quad \#\Big(\textstyle\prod_{e \mid d'_p} T_e\Big)(\mathbb{F}_p) \text{ is } B\text{-smooth},$$

a smoothness condition on the $\mathbb{F}_p$-points of a torus of dimension

$$\dim \prod_{e \mid d'_p} T_e \;=\; \sum_{e \mid d'_p} \varphi(e) \;=\; d'_p \;=\; \begin{cases} d_p/2, & d_p \text{ even}, \\ d_p, & d_p \text{ odd}. \end{cases}$$

Moreover:

(a) $d'_p \geq 2$ *for every* $p \mid n$*: the burden is never a single one-dimensional torus. (Indeed* $d'_p = 1$ *would force* $d_p \in \{1, 2\}$*, contradicting* $d_p \geq 3$ *from Lemma 3.1(iii).)*

(b) $d'_p = 2$ *iff* $d_p = 4$*, in which case the level set is* $\{1, 2\}$ *and the condition is:* $p-1 = \#T_1(\mathbb{F}_p)$ *and* $p+1 = \#T_2(\mathbb{F}_p)$ *both B-smooth — the Carmichael + Lucas–Carmichael* double condition*, i.e. twin-smoothness of* $\left(\frac{p-1}{2}, \frac{p+1}{2}\right)$*.*

(c) $d_p = 4$ *is realizable iff* $4 \mid r - 1$*; hence for* $r \equiv 1 \pmod 4$ *the minimal burden is* 2*, identical in structure to the* $r = 5$ *case, while for* $r \equiv 3 \pmod 4$ *every admissible* d_p *gives burden* ≥ 3*.*

(d) The minimal burden as a function of r *is*

$$\text{burden}(r) \;=\; \min_{\substack{d \mid r-1 \\ d \geq 3}} \begin{cases} d/2, & d \text{ even} \\ d, & d \text{ odd} \end{cases}$$

which reproduces the measured A11 table for all eleven moduli tested there (Section 4).

Proof. The support statement is (5) plus Proposition 3.5; the dimension computation is $\sum_{e \mid m} \varphi(e) = m$ with $m = d'_p$, together with $\dim T_e = \varphi(e)$ and $\#T_e(\mathbb{F}_p) = \Phi_e(p)$ (Section 2). (a): $d_p \geq 3$ by Lemma 3.1(iii); $d'_p = 1$ requires $d_p \leq 2$. (b): $d'_p = 2$ with d_p even means $d_p = 4$ and levels $\{e : e \mid 2\} = \{1, 2\}$; $d'_p = 2$ with d_p odd is impossible ($d_p = 2$ is even). With p odd, B-smoothness of $p \mp 1 = 2m, 2(m+1)$ is twin-smoothness of the pair $(m, m+1)$ up to the factor 2. (c): $d_p = 4$ requires $4 \mid \#(\mathbb{Z}/r)^\times = r - 1$; conversely if $4 \mid r - 1$ the cyclic group $(\mathbb{Z}/r)^\times$ has elements of order 4, and primes p in those residue classes exist (Dirichlet), with $\text{ord}_r(n) = 4 \mid d_p$ satisfiable. For $r \equiv 3 \pmod 4$: even $d \mid r - 1$ satisfy $d \equiv 2 \pmod 4$, so even $d \geq 3$ means $d \geq 6$, burden $d/2 \geq 3$; odd $d \geq 3$ gives burden $d \geq 3$. (d) is the minimization of d'_p over admissible d_p. □

Corollary 3.7 (structural closure of option (c))**.** *For every AKS modulus* r*, the L–P mechanism requires each pool prime to satisfy a smoothness condition on the point-orders of a torus of dimension at least* 2*; no modulus reduces the requirement to a single one-dimensional torus (a single condition of the shape "*$p \pm 1$ *smooth"). The minimum, dimension exactly* 2*, is achieved precisely by the split/nonsplit pair* $\{T_1, T_2\}$ *— the twin-smooth double condition — and only for* $r \equiv 1 \pmod 4$*. The double condition measured empirically in A11 is therefore intrinsic: it is the unique minimizer of a dimension count, not an artifact of* $r = 5$*.*

Remark 3.8 (cost interpretation). Under the standard Dickman heuristic, B-smoothness of $\Phi_e(p)$, a quantity of size $p^{\varphi(e)}$, has cost governed by $u = \varphi(e) \log p / \log B$; total burden across the level set is governed by $\sum_{e \mid d'} \varphi(e) = d'$. Burden 2 is the twin-smooth regime this project has harvested at scale; burden ≥ 3 requires smooth values of $\Phi_e(p) \sim p^{\varphi(e)}$ with $\varphi(e) \geq 2$, the same quantities whose *non*-smoothness underpins torus-based cryptography [9]. This is the structural content of A11's "every other r is strictly worse".

Remark 3.9 (what remains conditional). Lemmas 3.1 and 3.2, the support bound (5), and parts (a)–(d) of Theorem 3.6 as statements about the support of $\rho_r(p)$ are unconditional. The conditional ingredient is generic sharpness (Proposition 3.5): a harvest could in principle target the thin set of primes where $\rho_r(p)$ collapses below (5). Such primes have heuristic density $O(1/\ell)$ per large prime ℓ avoided — i.e. the "collapsed" pool is far thinner than the twin-smooth pool itself — so this escape is uneconomical, but it is recorded here as the localized gap per gate AG-G1 (`doc/arith-geom-plan.md`).

4 Numerical validation

`code/ag2_torus_check.py` (deterministic; output in `data/generated/ag2_torus_check.json`) verifies, over 67 sample primes spanning $(r, d) \in \{(5,4),(13,4),(41,4),(7,3),(7,6),(19,3),(11,5),(23,11)\}$:

check	statement	result
V1	Frobenius matching (Lemma 3.1(i))	67/67
V2	collapse identity $(X-1)^{p^{d/2}-1} = -X^{-1}$ (Lemma 3.2)	36/36
V3	exact $\rho_r(p)$ computed within bound (5)	67/67
V4	large primes of levels $e \mid d'$ divide $\rho_r(p)$ (Prop. 3.5)	37/37
V5	top level $e = d'$ genuinely needed in $\rho_r(p)$	67/67
V6	burden formula (Thm. 3.6(d)) vs. A11 table	11/11

V4's 0 exceptions are consistent with the $1/\ell$ model (≈ 0.19 misses expected over this sample). V6 reproduces, for $r \in \{5,7,11,13,17,19,23,29,31,37,41\}$, both the burden value and the minimizing d of the A11 measurement (`data/FINDINGS.md`, 2026-08-24).

5 A quantified heuristic for Popovych's conjecture

Popovych's conjecture [4] strengthens the hypothesis of Agrawal's to the simultaneous pair

$$(X-1)^n \equiv X^n - 1 \quad \text{and} \quad (X+2)^n \equiv X^n + 2 \pmod{n,\, X^r - 1}, \tag{6}$$

with the same conclusion. Beyond the Primaboinca search range ($n < 10^{17}$) there is no published evidence in either direction. This section quantifies what the second section costs the L–P mechanism — the only known counterexample engine — and finds that it destroys exactly the structure that made Agrawal's conjecture attackable: the inversion collapse of Lemma 3.2, and with it the twin-smooth (burden-2) regime.

Throughout, $r \neq 3$ is prime and $p \nmid 3(2^r+1)$; the finitely many excluded p are degenerate ($p = 3$ makes $-2 \equiv 1$, so the two sections of (6) *coincide* mod 3; $p \mid (2^r+1)/3$ makes $X+2$ a zero divisor in some component).

Lemma 5.1 (the level-one component is base-3 Fermat)**.** *The $X = 1$ component of the second congruence in (6) is $3^n \equiv 3 \pmod n$. In particular it is automatic for any Carmichael-type n ($p-1 \mid n-1$ for all $p \mid n$), hence free for the L–P construction; but unlike the first section, whose $X = 1$ component is $0 = 0$, the second section is already nontrivial at level one.*

Proof. Evaluate (6) at $X = 1$: $(1+2)^n \equiv 1 + 2 \pmod n$. □

Lemma 5.2 (global independence)**.** *In $\mathbb{Z}[\zeta_r]$ the elements $\zeta - 1$ and $\zeta + 2$ have principal ideals with disjoint prime support: $N(\zeta - 1) = \Phi_r(1) = r$ while $N(\zeta + 2) = \Phi_r(-2) = (2^r+1)/3$, and*

$\gcd\big(r,\,2^r+1\big) = \gcd(r,3) = 1$. *Since neither is a unit,* $\zeta-1$ *and* $\zeta+2$ *generate a rank-2 subgroup of* $\mathbb{Q}(\zeta_r)^\times$ *modulo roots of unity: no multiplicative identity ties the order of one reduction to the order of the other.*

Proof. $X^r-1=(X-1)\Phi_r(X)$ gives $\Phi_r(-2)=((-2)^r-1)/(-3)=(2^r+1)/3$. By Fermat $2^r\equiv 2 \pmod r$, so $2^r+1\equiv 3 \pmod r$ and $\gcd(r,2^r+1)=\gcd(r,3)=1$ as $r\neq 3$. Disjoint support plus non-unit norms gives independence modulo torsion. □

Lemma 5.3 (trivial stabilizer: no collapse at any depth)**.** *For every* $\gamma\in(\mathbb{Z}/r)^\times$ *with* $\gamma\neq 1$*, the ratio* $\sigma_\gamma(\zeta+2)/(\zeta+2)=(\zeta^\gamma+2)/(\zeta+2)$ *is not a root of unity — it is not even a unit of* $\mathbb{Z}[\zeta_r]$*. Consequently the mechanism of Lemma 3.2 has no analogue for* $X+2$*: no relation* $(X+2)^{p^e-1}\in\mu_{2r}$ *holds for any* e *with* $p^e\not\equiv 1 \pmod r$*.*

Proof. Let $\mathfrak{q}$ be a prime of $\mathbb{Z}[\zeta_r]$ dividing $(\zeta+2)$, so $\zeta\equiv -2 \pmod{\mathfrak{q}}$ and the image of -2 in the residue field has multiplicative order r (it is the image of ζ). If $\mathfrak{q}$ also divided $(\zeta^\gamma+2)$ then $\zeta^\gamma\equiv -2\equiv\zeta$, so $\zeta^{\gamma-1}\equiv 1 \pmod{\mathfrak{q}}$, forcing $r\mid\gamma-1$, i.e. $\gamma=1$. So for $\gamma\neq 1$ the ratio has nonzero valuation at $\mathfrak{q}$ and is not a unit.

Contrast with the first section: $\sigma_{-1}(\zeta-1)/(\zeta-1)=-\zeta^{-1}\in\mu_{2r}$, available because the ideal $(\zeta-1)$ (the totally ramified prime above r) is fixed by the whole Galois group — geometrically, the zero locus $\{1\}$ of $X-1$ is Galois-stable, while the zero -2 of $X+2$ has trivial stabilizer. □

Corollary 5.4 (full-level support of the second section)**.** *Define* $\rho'_r(p)$ *as the multiplicative order of* $X+2$ *in* $(\mathbb{F}_p[X]/\Phi_r(X))^\times$*. Then* $\rho'_r(p)\mid p^d-1=\prod_{e\mid d}\Phi_e(p)$*, and — in the generic-sharpness sense of Proposition 3.5, now justified globally by Lemma 5.3 — its support includes* all *levels* $e\mid d$*, in particular the top level* $\Phi_d(p)$ *of dimension* $\varphi(d)$*, for even* d *the level the first section never sees.*

Theorem 5.5 (Popovych burden)**.** *Run the L–P mechanism for the pair (6): for each* $p\mid n$*,*

$$n\equiv p^{j_p} \pmod{\operatorname{lcm}\big(r,\ \rho_r(p),\ \rho'_r(p)\big)},\qquad \operatorname{ord}_r(n)>2.$$

Under generic sharpness for both sections, the combined support is all levels $e\mid d_p$*, so the smoothness burden at* p *is* $\sum_{e\mid d_p}\varphi(e)=d_p$ *for* every *parity of* d_p*. Hence*

$$\operatorname{burden}_{\mathrm{Pop}}(r)\ =\ \min\{\,d : d\mid r-1,\ d\geq 3\,\}\ \geq\ 3\qquad \textit{for every prime } r,$$

and:

(a) the global optimum is 3 *(attained iff* $3\mid r-1$*), versus* 2 *for the single section (Theorem 3.6);*

(b) at $r=5$ *— Popovych's own proposed modulus — and generally whenever* $4\mid r-1$ *and* $3\nmid r-1$*, the burden is* 4*: the pool primes must have* $p-1$*,* $p+1$ and p^2+1 *simultaneously smooth;*

(c) $\operatorname{burden}_{\mathrm{Pop}}(r)\geq\operatorname{burden}_{\mathrm{Agr}}(r)$ *for every* r*, strictly whenever* $\operatorname{burden}_{\mathrm{Agr}}(r)=2$ *(i.e. for every* $r\equiv 1 \pmod 4$*);*

(d) the twin-smooth regime (burden 2*) — the only regime this project found supply-feasible (gates G1/G2) — does not exist for the pair at any modulus.*

Proof. For odd d both sections are supported on all $e\mid d$ already (Section 3), total d; the pair ties with the single section there. For even d the first section collapses to $e\mid d/2$ (Lemma 3.2) but the second does not (Corollary 5.4), so the lcm spans all $e\mid d$, total $\sum_{e\mid d}\varphi(e)=d$ — the collapse is destroyed. Admissibility $d\geq 3$ is Lemma 3.1(iii), unchanged. Minimizing d over $d\mid r-1$, $d\geq 3$

gives the formula. (a): the least divisor ≥ 3 of $r-1$ is 3 iff $3 \mid r-1$. (b): if $3 \nmid r-1$ and $4 \mid r-1$ the least is 4, with levels $\{1,2,4\}$ and $\Phi_4(p) = p^2+1$. (c): $\mathrm{burden}_{\mathrm{Agr}}(r)$ minimizes d' where $d' \leq d$ always, so it is $\leq \mathrm{burden}_{\mathrm{Pop}}(r)$; when it equals 2 the Popovych value is ≥ 3 by the displayed bound. (d) is (a) plus Theorem 3.6(b). □

Proposition 5.6 (cost at the measured scale)**.** *At the frozen harvest spec of this project (`doc/harvest-spec.md`: pool primes $p \approx 51$ bits, measured supply/demand ratio 13.8, pool 2.5×10^5), the extra quadratic level of Theorem 5.5(b) is a pure Dickman tax on a ≈ 102-bit quantity: $\rho(u)$ with $u = 102/\log_2 B$, i.e.*

$$u \approx 9.9,\ \rho(u) \approx 3.9 \times 10^{-11} \quad (B = 1259), \qquad u \approx 7.5,\ \rho(u) \approx 1.8 \times 10^{-7} \quad (B' = 12590),$$

multiplying the measured ratio down to 5×10^{-10}–2.5×10^{-6} and the pool to 10^{-5}–0.05 elements: the pair-mechanism harvest is empty at any feasible scale. *Unlike $p \mp 1$, where partition-first harvesting lets one side be enumerated rather than tested (`data/FINDINGS.md`, A1), $\Phi_4(p)$ is pinned once p is, so no enumeration trick removes this tax. On the solver side the group acquires the level-4 components, so $\log_2 |G|$ grows and the identity density falls below the measured $\rho \approx 0.004$ of H7 pools: the N1/A10 obstruction for the pair strictly contains the one for the single section.*

Remark 5.7 (interpretation and scope). Theorem 5.5 and Proposition 5.6 are, to our knowledge, the first quantified evidence in either direction on Popovych's conjecture beyond exhaustive search: every known counterexample mechanism pays at least one more torus dimension against the pair than against the single section globally (3 vs. 2), and two more at the canonical $r = 5$ (4 vs. 2); the burden-2 regime that made the Agrawal attack supply-feasible does not exist for the pair. The caveats are those of Remarks 2.3 and 3.9, applied to both sections: this quantifies the failure of the only known engine, not the truth of the conjecture. Note also the char-3 degeneracy: modulo $p = 3$ the two sections coincide, so the pair constrains only $p > 3$ divisors.

5.1 Numerical validation

`code/ag5_popovych_check.py` (deterministic; output in `data/generated/ag5_popovych_check.json`) verifies, over 56 sample primes spanning the same (r, d) grid as Section 4:

check	statement	result
P1	Frobenius matching for $X+2$	56/56
P2	the $X-1$ collapse bound does *not* absorb $X+2$ (even d)	35/36
P3	top level $e = d$ present in $\rho'_r(p)$	55/56
P4	large primes of *all* levels $e \mid d$ divide $\rho'_r(p)$	47/47
P5	burden table: $\mathrm{burden}_{\mathrm{Pop}}$ vs. $\mathrm{burden}_{\mathrm{Agr}}$, 11 moduli	as stated
P6	$\gcd(r, 2^r+1) = 1$ (Lemma 5.2)	11/11
P7	Dickman cost table (Proposition 5.6)	as stated

The single P2/P3 exception is $r = 13$, $p = 5$: there $\Phi_4(5) = 26 = 2 \cdot 13$ contains no prime outside $2r$ at all, so the top level is vacuous for this one tiny prime — a small-prime degeneracy, not a collapse event; it is consistent with (indeed predicted by) the generic model, under which exceptions have probability $\sim 1/\ell$ per large prime ℓ and ≈ 0.28 misses were expected over this sample. The burden deltas of P5: $+2$ at $r \in \{5, 17, 29, 41\}$, $+1$ at $r \in \{13, 37\}$, 0 at the $r \equiv 3 \pmod 4$ moduli (where the Agrawal burden was already ≥ 3 and both conjectures share full depth).

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